

Wavelet Theory

$\psi(t)$: Wavelet

i) $\int_{-\infty}^{\infty} \psi(t) dt = 0$ 'oscillatory'

ii) $\int_{-\infty}^{\infty} |\psi(t)|^2 dt < \infty$ finite energy.

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right)$$

Cont. Wavelet Transform & its Inverse

$$f(t) \longleftrightarrow W(a, b) \quad a = \text{scale}, b = \text{shift}$$

$$W(a, b) = \int_{-\infty}^{\infty} f(t) \frac{1}{\sqrt{a}} \psi^*\left(\frac{t-b}{a}\right) dt \quad \begin{matrix} 0 < a < \infty \\ -\infty < b < \infty \end{matrix}$$

$$f(t) = \frac{1}{C} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W(a, b) \psi_{a,b}(t) \frac{da db}{|a|^2}$$

$$C \equiv \int_{-\infty}^{\infty} \frac{|\psi(\eta)|^2}{|\eta|^2} d\eta \quad 0 < C < \infty.$$

Dyadic Wavelet Decomposition

Let us sample 'a' at $a = 2^{-j}$ $-\infty < j < \infty$
and 'b' at $b = 2^{-j}k$

Then $\psi_{j,k}(t) = 2^{j/2} \psi(2^j t - k)$

Now $W(a, b)$ at $a = 2^{-j}$
 $b = k 2^{-j}$

$W(2^{-j}, k 2^{-j})$ will be denoted as
 "d_{jk}"

i.e. d_{jk} = Coef of $2^{j/2} \psi(2^j t - k)$

Dyadic Wavelet expansion (decomposition)
 of a function $f(t)$ is now given by

$$f(t) = \sum_j \sum_k d_{jk} \psi_{jk}(t)$$

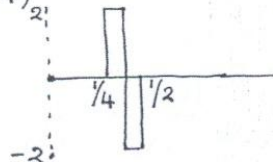
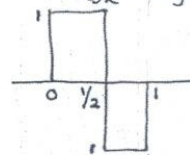
If ψ_{jk} are orthogonal
 $d_{jk} = \langle f(t), \psi_{jk}^* \rangle$

Haar Wavelet

$$\psi(t) = \psi_{j,0}(t)$$

$$\psi_{2,1}(t) = 2^{1/2} \psi(2^2 t - 1)$$

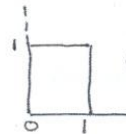
$$\text{Energy} = 2^2 \cdot \frac{1}{4} = 1$$



Haar Scaling function

$$\phi(t)$$

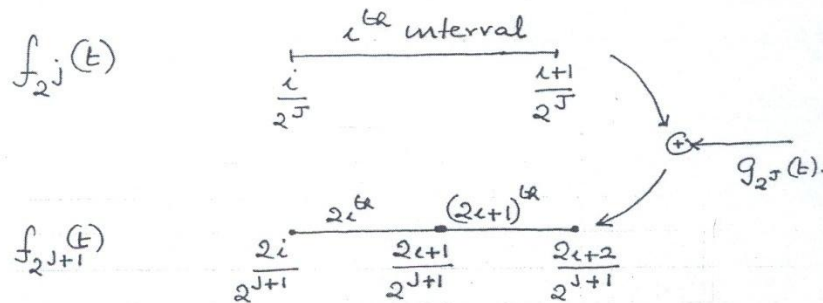
Haar Scaling function



$\phi(t)$ for Haar

Linear Piecewise Constant approx of a signal $f(t)$

This kind of approx ^{Corresponds} leads to Haar Wavelet/
Scaling function application to $f(t)$



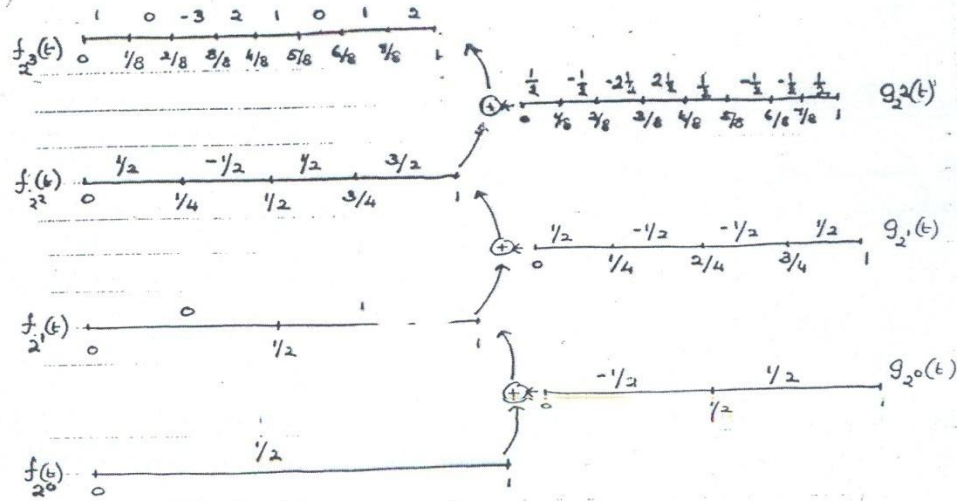
$g_{2^J}(t)$ = detail signal at level J

$f_{2^J}(t)$ = approx at level J of the signal $f(t)$

$$g_{2^J}(t) = f_{2^{J+1}}(t) - f_{2^J}(t)$$

$$\lim_{J \rightarrow \infty} f_{2^J}(t) = f(t) = \sum_{J=-\infty}^{\infty} g_{2^J}(t)$$

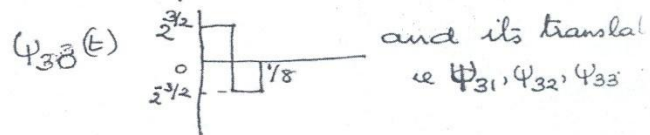
W-4



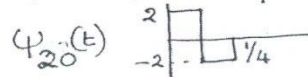
$f_{2^J}(t)$ Can be expressed as linear combination of ψ_{Jk}

$g_{2^J}(t)$ " " " Φ_{Jk}

Eg: $g_{2^3}(t)$ Can be expressed as a l.c of



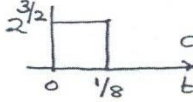
$g_{2^2}(t)$ → Can be expressed as l.c of $\psi_{20}(t)$ and its translates i.e. $\psi_{21}, \psi_{22}, \psi_{23}$



~~W-5~~
W-49

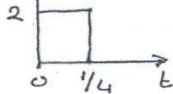
Similarly

$f_{23}(t)$ can be expressed as a l.c of

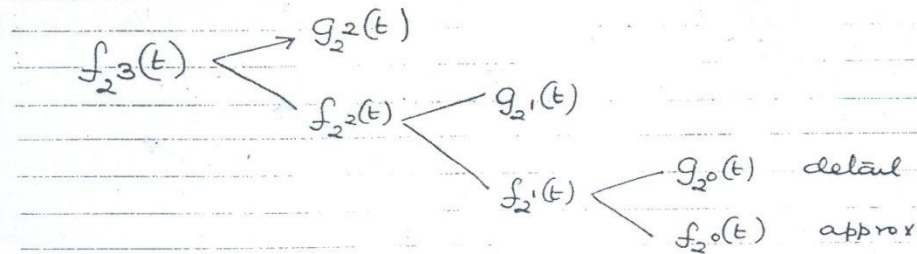
$\phi_{30}(t)$  and its translates $\phi_{31}, \phi_{32}, \dots$

$$\phi_{30}(t) = 2^{3/2} \phi(2^3 t)$$

$f_{22}(t)$ can be expressed as a l.c of

$\phi_{20}(t)$  and its translates $\phi_{21}, \phi_{22}, \phi_{23}, \dots$

Decomposition



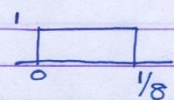
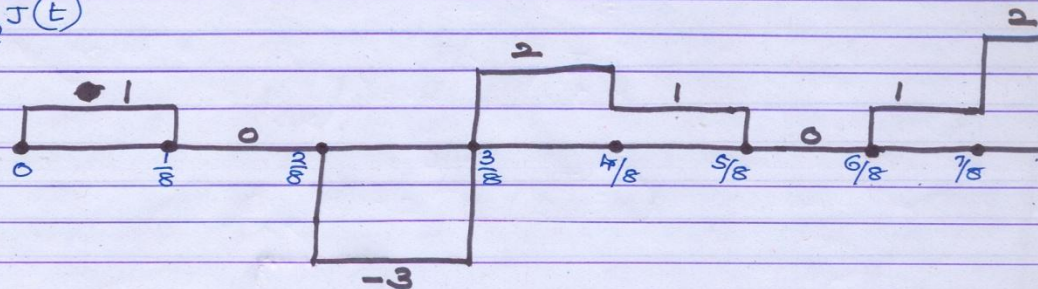
$$f_{23}(t) = g_{22}(t) + g_{21}(t) + g_{20}(t) + f_{20}(t)$$

$$\phi_{jk}(t) = 2^{j/2} \phi(2^j t - k)$$

$$\psi_{jk}(t) = 2^{j/2} \psi(2^j t - k)$$

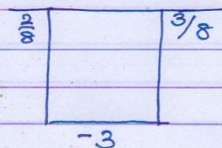
N-5

$\phi_{2^j}(t)$



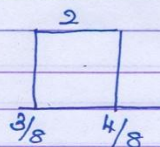
$$= \phi(2^3 t - 0) = 2^{-3/2} \cdot 2^{3/2} \phi(2^3 t - 0) = 2^{-3/2} \phi_{30}(t)$$

$$= \frac{1}{2\sqrt{2}} \phi_{30}(t)$$



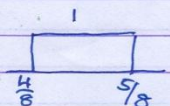
$$= -3 \phi(2^3 t - 2) = -3 \cdot 2^{-3/2} \cdot 2^{3/2} \phi(2^3 t - 2)$$

$$= -\frac{3}{2\sqrt{2}} \phi_{32}(t)$$



$$= 2 \phi(2^3 t - 3) = 2 \cdot 2^{-3/2} \cdot 2^{3/2} \phi(2^3 t - 3)$$

$$= \frac{1}{\sqrt{2}} \phi_{32}(t)$$



$$= 2^{-3/2} \cdot 2^{3/2} \phi(2^3 t - 3) = \frac{1}{2\sqrt{2}} \phi_{33}(t)$$

$$\frac{1}{2\sqrt{2}}, 0, -\frac{3}{2\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}, 0, \frac{1}{2\sqrt{2}}, \frac{1}{\sqrt{2}}$$

4-6

(4)

$$f_{23}(t) = \frac{1}{2\sqrt{2}} \phi_{30}(t) + 0 \cdot \phi_{31}(t) + \left(\frac{-3}{2\sqrt{2}}\right) \phi_{32}(t)$$

$$+ \frac{1}{2\sqrt{2}} \phi_{33}(t) + \frac{1}{2\sqrt{2}} \phi_{34}(t) + 0 \cdot \phi_{35}(t)$$

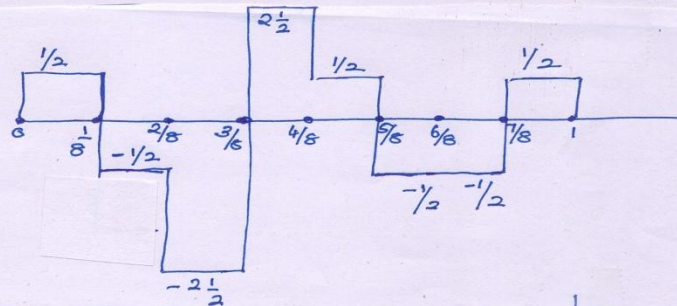
$$+ \frac{1}{2\sqrt{2}} \phi_{36}(t) + \frac{1}{\sqrt{2}} \phi_{37}(t)$$

$$= \left\{ \frac{1}{2\sqrt{2}}, 0, \frac{3}{2\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}, 0, \frac{1}{2\sqrt{2}}, \frac{1}{\sqrt{2}} \right\}$$

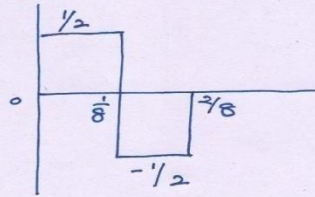
$$f_{23}(t) = \sum_k c_{3k} \phi_{3k}$$

$$f_{2J}(t) = \sum_k c_{jk} \phi_{jk}$$

$$g_2(t) =$$

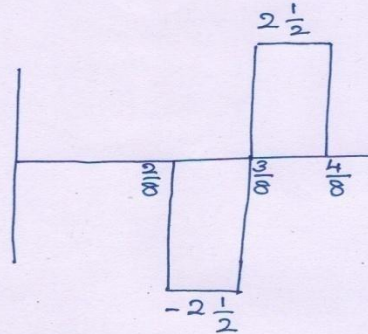


N-7 B



$$\frac{1}{2} \psi(4t-0) = \frac{1}{4} \cdot 2^1 \psi(2^2 t - 0)$$

$$= \frac{1}{4} \psi_{20}(t)$$



$$= -\frac{5}{4} \cdot 2^1 \psi(2^2 t - 1)$$

$$= -\frac{5}{4} \psi_{21}(t)$$

Note

$$f_{2^J}(t) = \text{l.c of } \phi_{jk}(t) \quad k \in \mathbb{Z}, \dots, 0, 1, 2, \dots$$

$$g_{2^J}(t) = \text{l.c of } \psi_{jk}(t) \quad \text{''} \quad \text{''}$$

Example

$$f_{2^3}(t) = \sum_k c_{3k} \phi_{3k} \left\{ \begin{array}{l} g_{2^2}(t) = \sum_k c_{2k} \psi_{2k}(t) \\ f_{2^2}(t) = \sum_k c_{2k} \phi_{2k}(t) \end{array} \right. \left\{ \begin{array}{l} g_2(t) \\ f_2(t) \end{array} \right.$$

$$f_2(t) = [d_{20}\psi_{20} + d_{21}\psi_{21} + d_{22}\psi_{22} + d_{23}\psi_{23}] \\ + [d_{10}\psi_{10} + d_{11}\psi_{11}] + [d_{00}\psi_{00}] + c_{00}\phi_{00}$$

$$[d_{20}, d_{21}, d_{22}, d_{23}] = [\frac{1}{4}, -\frac{5}{4}, \frac{1}{4}, -\frac{1}{4}]$$

$$[d_{10}, d_{11}] = [\frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}]$$

$$d_{00} = -\frac{1}{2} \quad c_{00} = \frac{1}{2}$$

$$\begin{aligned} f_2 \rightarrow & \left[\frac{1}{2\sqrt{2}}, 0, -\frac{3}{2\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}, 0, \frac{1}{2\sqrt{2}}, \frac{1}{\sqrt{2}} \right] \begin{bmatrix} [\frac{1}{4}, -\frac{5}{4}, \frac{1}{4}, -\frac{1}{4}] \{\psi_{2k}\} \\ [\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}] \{\psi_{1k}\} \\ [0, \frac{1}{\sqrt{2}}] \{\psi_{0k}\} \end{bmatrix} \\ & \begin{bmatrix} \{\phi_{2k}\} \\ \{\phi_{1k}\} \\ \{\phi_{0k}\} \end{bmatrix} \end{aligned}$$

$$\phi_{jk} = 2^{j/2} \phi(2t-k)$$

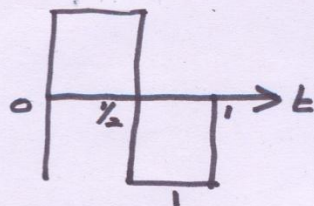
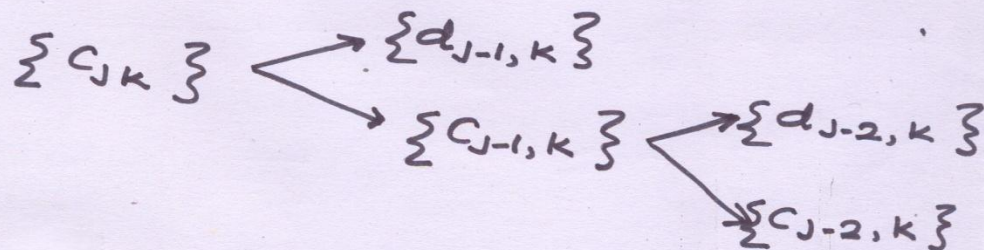
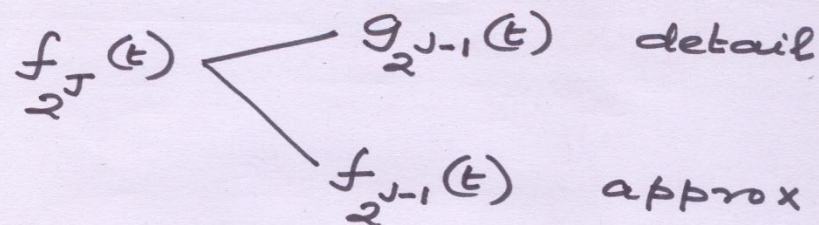
$$\psi_{jk} = 2^{j/2} \psi(2t-k)$$

$$[c_{30}, c_{31}, c_{32}, c_{33}, c_{34}, c_{35}, c_{36}, c_{37}, c_{38}] \begin{bmatrix} G \\ H \\ H \end{bmatrix} \begin{bmatrix} G \\ G \\ H \end{bmatrix} \begin{bmatrix} [d_{20}, d_{21}, d_{22}, d_{23}] \\ [d_{10}, d_{11}] \\ [d_{00}] \end{bmatrix} \begin{bmatrix} [c_{00}] \end{bmatrix}$$

$$\phi(t) = \sum R(t) \sqrt{2} \phi(2t-k)$$

W9 4

One level Decomposition



$$\begin{aligned}\phi(t) &= \phi(2t) - \phi(2t-1) \\ &= \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \phi(2t) - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \phi(2t-1)\end{aligned}$$

$$\phi_{00}(t) = \frac{1}{\sqrt{2}} \phi_{10}(t) - \frac{1}{\sqrt{2}} \phi_{11}(t)$$

$$\phi_{00}(t) = \frac{1}{\sqrt{2}} \phi_{10}(t) + \frac{1}{\sqrt{2}} \phi_{11}(t)$$

~~(5)~~ W-10

$$1) \psi(t) = \sum_k g(k) \phi_{1k}(t)$$

$$2) \phi(t) = \sum_k h(k) \phi_{1k}(t) = \frac{1}{\sqrt{2}} \phi_{10}(t) + \frac{1}{\sqrt{2}} \phi_{11}(t)$$

$$h(0) = h(1) = \frac{1}{\sqrt{2}}$$

$$\begin{aligned} \psi(t) &= \sum_k g(k) \phi_{1k}(t) \\ &= \frac{1}{\sqrt{2}} \phi_{10}(t) - \frac{1}{\sqrt{2}} \phi_{11}(t) \end{aligned}$$

$$g(0) = \frac{1}{\sqrt{2}}, \quad g(1) = -\frac{1}{\sqrt{2}}$$

One Can Show

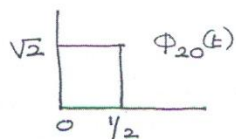
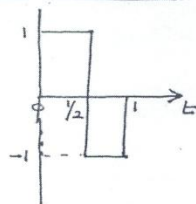
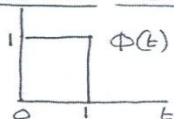
$$g(k) = (-1)^k h(1-k)$$

$$g(0) = h(1) = \frac{1}{\sqrt{2}}$$

$$g(1) = -h(0) = -\frac{1}{\sqrt{2}}$$

(1) + (2) $\left. \begin{array}{l} \text{Dilation} \\ \text{Two Scale} \end{array} \right\} \text{eqn.}$

Relation between Haar Wavelet & Scaling function



$$\phi(t) = \phi_{00}(t) = \frac{1}{\sqrt{2}} \phi_{20}(t) + \frac{1}{\sqrt{2}} \phi_{21}(t)$$

$$(A) \quad \phi(t) = \sum_k h(k) \phi_{2^k}(t) \quad \text{(in general)} \quad \leftarrow \quad h(0) = h(1) = 1/\sqrt{2}$$

$$\psi(t) = \frac{1}{\sqrt{2}} \phi_{20}(t) - \frac{1}{\sqrt{2}} \phi_{21}(t)$$

$$(B) \quad \psi(t) = \sum_k g(k) \phi_{2^k}(t) \quad \text{here } k=0, 1 \quad \text{(in general)}$$

One can show $g(k) = (-1)^k h(-k+1)$

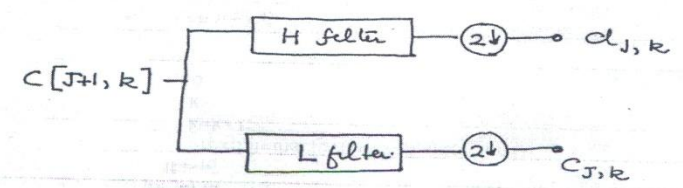
eg: $g(0) = h(1) = 1/\sqrt{2}$

$g(1) = -1 \times h(0) = -1/\sqrt{2}$

A & B are called Dilation or two scale equation

W-12
W-8

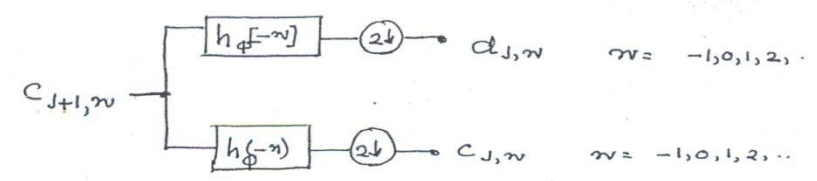
$$C_{(J+1,k)} \begin{cases} d_{J,k} & k = \dots, 0, 1, 2 \\ c_{J,k} & k = -1, 0, 1, \dots \end{cases}$$



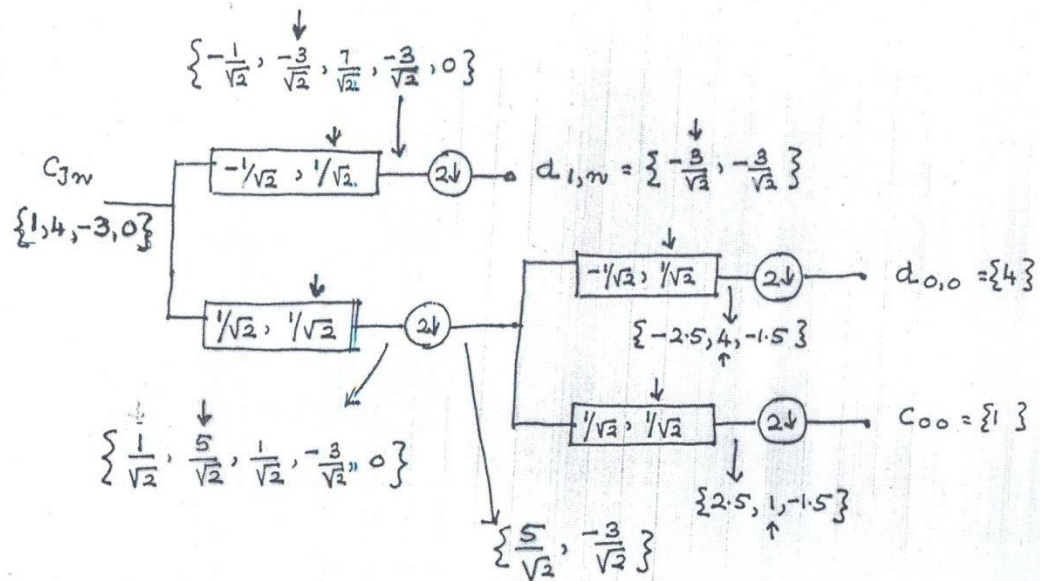
$$\begin{aligned} \text{H filter} &: h_{\psi}(-n) & h_{\phi}[n] &= \{h_0, h_1\} \\ \text{L filter} &: h_{\phi}(-n) & h_{\psi}[n] &= \{g_0, g_1\} \end{aligned}$$

$$d_{(J,k)} = \sum_m h_{\psi}(m-2k) c_{(J+1,k)}$$

$$c_{(J,k)} = \sum_m h_{\phi}(m-2k) c_{(J+1,k)}$$

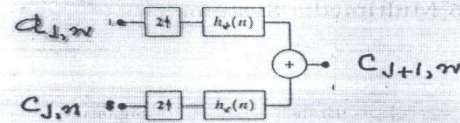


N-9

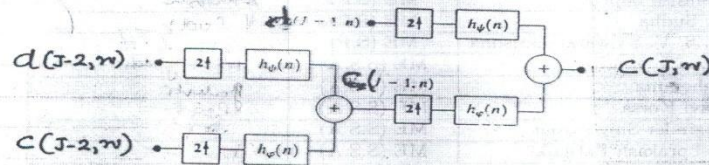


N-13

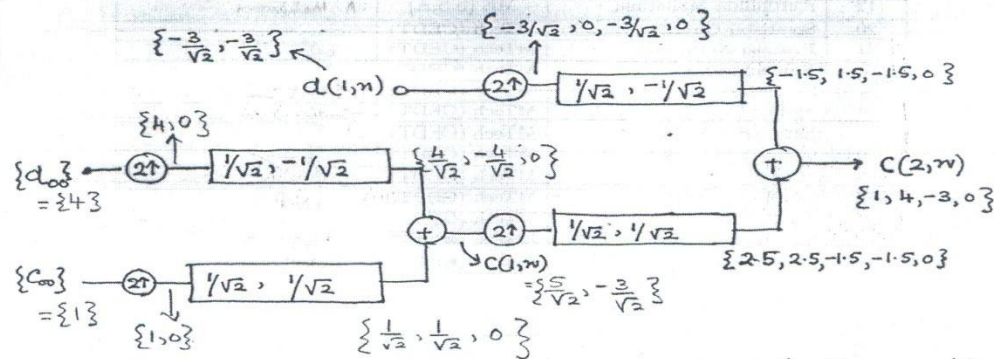
W-14



Synthesis filter.



2-stage synthesis filter.



2-scale Inverse Wavelet Transform.

FIGURE 7.15 An FWT analysis bank

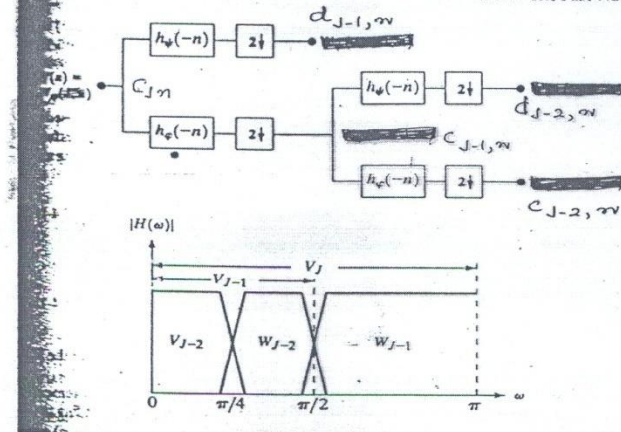
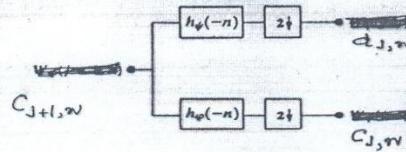


FIGURE 7.16 (a) A two-stage or two-scale FWT analysis bank and (b) its frequency splitting characteristics

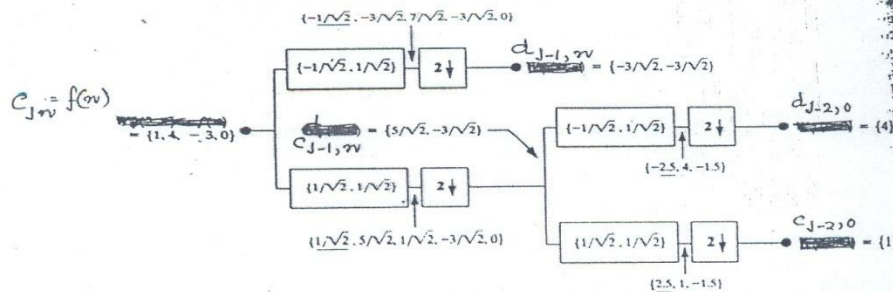


FIGURE 7.17 Computing a two-scale fast wavelet transform of sequence $\{1, 4, -3, 0\}$ using Haar scaling and wavelet vectors.

$$\begin{array}{c} \downarrow \\ -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \\ 0 \quad -3 \quad 4 \quad 1 \end{array}$$

$$\frac{1}{\sqrt{2}}$$

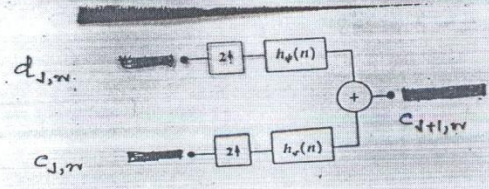


FIGURE 7.18 The FWT⁻¹ synthesis filter bank.

FIGURE 7.19 A two-stage or two-scale FWT⁻¹ synthesis bank.

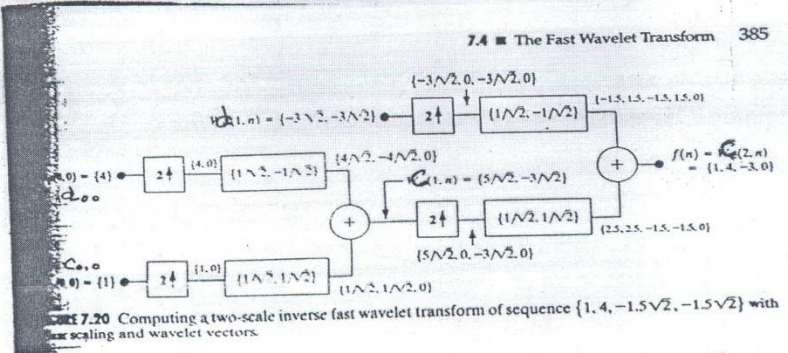
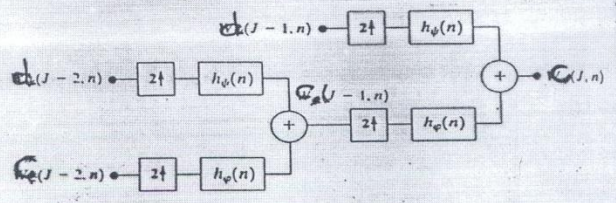


FIGURE 7.20 Computing a two-scale inverse fast wavelet transform of sequence $\{1, 4, -1.5\sqrt{2}, -1.5\sqrt{2}\}$ with scaling and wavelet vectors.